



Regulation at full load and ____ pf. Lag

$$BC = I_a \times X_{al} = \text{____} V (\text{from Potier triangle})$$

Regulation at full load and ____ pf. Lag

$$X_{al} = \text{____} \Omega$$

$$I_{far} = AB$$

$$\bar{V} = V \angle 0^\circ = \text{____} V$$

$$\cos \phi = \text{____} \text{ lag}$$

$$\bar{I} = I \angle (-\phi)^\circ = \text{____} A$$

$$\bar{E}_r = V \angle 0^\circ + I \angle (-\phi)^\circ \times (R_a + jX_{al}) = \text{____} V = E_r \angle \sigma^\circ$$

Refer OCC and find I_{fr} corresponding to E_r .

$$\bar{I}_{fr} = I_{fr} \angle (\sigma + 90)^\circ = \text{____} A$$

$$\bar{I}_{far} = I_{far} \angle (-\phi + 180)^\circ = \text{____} A \quad \text{for lag } (\bar{I}_{far} = I_{far} \angle (+\phi + 180^\circ) \text{ for lead})$$

$$\bar{I}_f = \bar{I}_{fr} + \bar{I}_{far} = \text{____} A = I_f \angle (90 + \delta)^\circ$$

$$\text{Hence, } I_f = \text{____} A, \delta = \text{____}^\circ$$

Refer OCC and find E_f corresponding to I_f

$$\bar{E}_f = E_f \angle \delta^\circ = \text{____} V$$

$$\% \text{ regulation} = \frac{E_f - V}{V} \times 100 \% = \text{____} \%$$

TABULATION - Potier method

	Power factor	$\bar{I}$	$\bar{E}_r$	$\bar{I}_{fr}$	$\bar{I}_{far}$	$\bar{I}_f$	$\bar{E}_f$	% regulation
1	0 lag							
2	0.2 lag							
3	0.4 lag							
4	0.6 lag							
5	0.8 lag							
6	1							